ACTIVE FILTER DESIGN & IMPLEMENTATION

Sallen-Key Op-Amp Filters: Theory, Simulation & Hardware Lab

Low-Pass · High-Pass · Band-Pass (Butterworth Response)

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Platforms used: Tinkercad · Multisim · LabVIEW · MATLAB · Hardware Lab

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1. Introduction & Objectives

This project tends to investigatethe design, simulation, and hardware implementation of second-order active filters built around operational amplifiers, using the Sallen-Key topology with a Butterworth (maximally-flat) response. The same core design is realized across five environments — Tinkercad, Multisim, MATLAB, and physical hardware — to compare theoretical predictions against simulated and measured behavior.

1.1 Objectives

- Derive and apply the design equations for Sallen-Key low-pass, high-pass, and band-pass active filters.
- Select standard resistor/capacitor values that realize a target cutoff frequency and Butterworth Q.
- Simulate each filter in Tinkercad, Multisim, and MATLAB, and process/measure frequency response.
- Build and test each filter on a breadboard using real op-amps.
- Compare theoretical, simulated, and measured frequency response; explain discrepancies using component tolerance and op-amp non-idealities.

1.2 Equipment & Software

- Breadboard, dual power supply (±12 V), function generator, oscilloscope, Digital Multi-meter
- Op-amp: TL082 / TL072 (or LM741 for slew-rate comparison)
- Resistors and capacitors per design tables (Section 3)
- Tinkercad Circuits, NI Multisim, NI LabVIEW, MATLAB (Control System / Signal Processing Toolbox)

2. Theoretical Background

2.1 Active vs. Passive Filters

Active filters use an amplifying element (the op-amp) in addition to resistors and capacitors. This avoids the signal loss of purely passive RC/RLC filters, provides high input impedance and low output impedance (so the filter does not load the source or following stage), and can supply voltage gain in the passband.

2.2 Filter Order and Roll-off

A first-order filter attenuates at 20 dB/decade beyond cutoff; a second-order filter (as used in this lab) attenuates at 40 dB/decade, giving a sharper transition between passband and stopband.

2.3 Filter Approximations

- Butterworth: maximally flat passband, moderate roll-off — used in this lab.
- Chebyshev: passband ripple, sharper roll-off than Butterworth for the same order.
- Bessel: linear phase response, gentler roll-off — preserves waveform shape.

2.4 Sallen-Key Transfer Function

For the low-pass configuration:

$$H_{(s)} = \frac{\omega_0^2}{(s^2+\left(\frac{\omega_0}{Q}\right).s+\omega_0^2)} \qquad f_0: cutoff\ frequency$$

$$f_0 = \frac{1}{(2\pi.\sqrt{R_1.R_2.C_1.C_2})}$$

Q: Quality Factor
Series RCL: $Q = \frac{\omega_0 L}{R}$

Parallel RCL: $Q = \omega_0 RC$

ζ: Damping ratio; It is a dimensionless measure that describes how oscillations in a system decay after a disturbance. If ζ<1, the system is underdamped and will oscillate.

$$\zeta = \frac{1}{2Q}$$

For the equal-component design (R1 = R2 = R, C1 = C2 = C) used in this project, Q is set independently by the non-inverting gain K of the op-amp stage:

$$K = 1 + \frac{R_f}{R_g} \qquad Q = \frac{1}{3-k}$$

Butterworth $\Rightarrow Q = 0.707 \Rightarrow K = 1.586$

2.5 Op-Amp Non-Idealities

- Gain-Bandwidth Product (GBW): [1]limits the usable frequency range — closed-loop gain falls off once $f\ approaches \frac{GBW}{K}$.
- Slew Rate[2]: limits how fast V_{out} can change; large-amplitude high-frequency signals become triangular (distorted) instead of sinusoidal.
- Input offset voltage / bias current: introduce small DC errors, mainly relevant at very low frequency or high-impedance designs.

2.6 Band-Pass by Cascading

A band-pass response is realized here by cascading a high-pass stage (sets the lower cutoff) with a low-pass stage (sets the upper cutoff), each independently designed using the equations above.

[1] Gain-Bandwidth Product. In active electronic filters, it specifies the maximum frequency at which an operational amplifier can provide useful gain. Because filters rely heavily on amplifiers to shape signals, an inadequate GBW causes degraded cutoff accuracy, unwanted peaking in the passband, and phase shifts.

[2] In active filters, slew rate is the maximum speed at which the circuit's output voltage can change. Measured in volts per microsecond (V/µs), it defines the filter's ability to process fast, high-amplitude signals without distorting them. If an input signal demands faster voltage changes than the limit, the output will round off.

3. Design Calculations & Component Values

Target center design frequency: 1 kHz (LPF/HPF); passband ≈ 300 Hz – 3 kHz (BPF). C = 10 nF chosen as the reference capacitor for all stages; R solved from f0 = 1/(2πRC) and rounded to nearest standard (E24) value.

Assume $C_1 = C_2 = 10nF$

Using cutoff frequency equation :

$$f_0 = \frac{1}{2\,\pi RC} \Longrightarrow \quad R = \frac{1}{2\pi F_0 C} = \frac{1}{2\pi.1000.(10.10^{-9})} = 15.915\,\Omega \cong 16\,\Omega$$

$$\because\ 16\,k\Omega\ isn't\ exactly\ 15{,}915\,\Omega \Longrightarrow the\ real\ cutoff\ shifts\ slightly$$

$$f_0 = \frac{1}{2(16{,}000)(10.10^{-9})} = \frac{1}{2\pi(1.6.10^{-4})} = \frac{1}{1.005309649.10^{-3}} = 994.7183943$$
$$\cong\ 994.7\ Hz$$

For Butterworth, we derived (Section 2.6): $Q = 0.707 \Rightarrow K = 1.586$

$$K = 1 + \frac{R_f}{R_g} \Rightarrow \frac{R_f}{R_g} = K - 1 = 0.586$$

Lets take$R_g = 10k\Omega$

$$R_f = 0.586\,.10{,}000 = 5{,}860\,\Omega$$

Nearest standard E24 value: 5.6 kΩ small rounding, ~4.4% off the exact 5.86 kΩ target

this is why the earlier response flagged it as introducing a small Q_{error} ~1.8%, rather than an f_0 error.

3.1 Low-Pass Filter — Component Table

Parameter	Value
R1 = R2	16 kΩ
C1 = C2	10 nF
R_g	10 kΩ
R_f	5.6 kΩ
Op-amp	LM741
Supply	±12 V
Target f_0	1000 Hz
Calculated actual f_0	994.7 Hz

3.2 High-Pass Filter — Component Table

Parameter	Value
C1 = C2	10 nF
R1 = R2	16 kΩ
Rg	10 kΩ
Rf	5.6 kΩ
Op-amp	LM741
Supply	±12 V
Target f0	1000 Hz
Calculated actual f0	994.7 Hz

3.3 Band-Pass Filter — Component Table

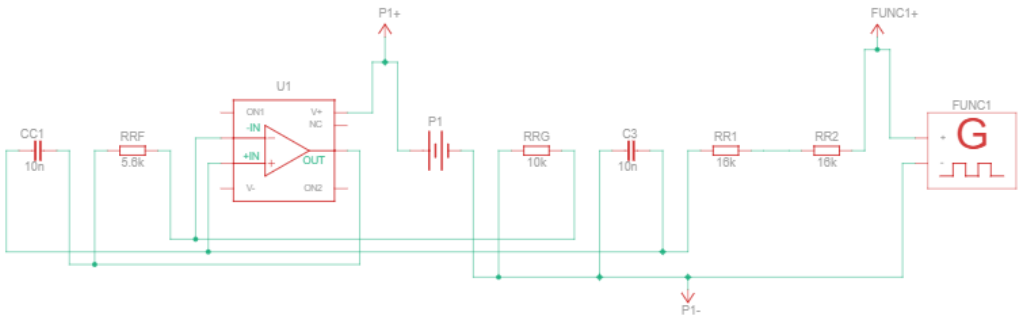
Parameter	Value
R_1	$10k\Omega$
C	10nF
L	26H
F_0	312
Op-amp	LM741

4. Experiment 1 — Sallen-Key Low-Pass Filter

4.1 Objective

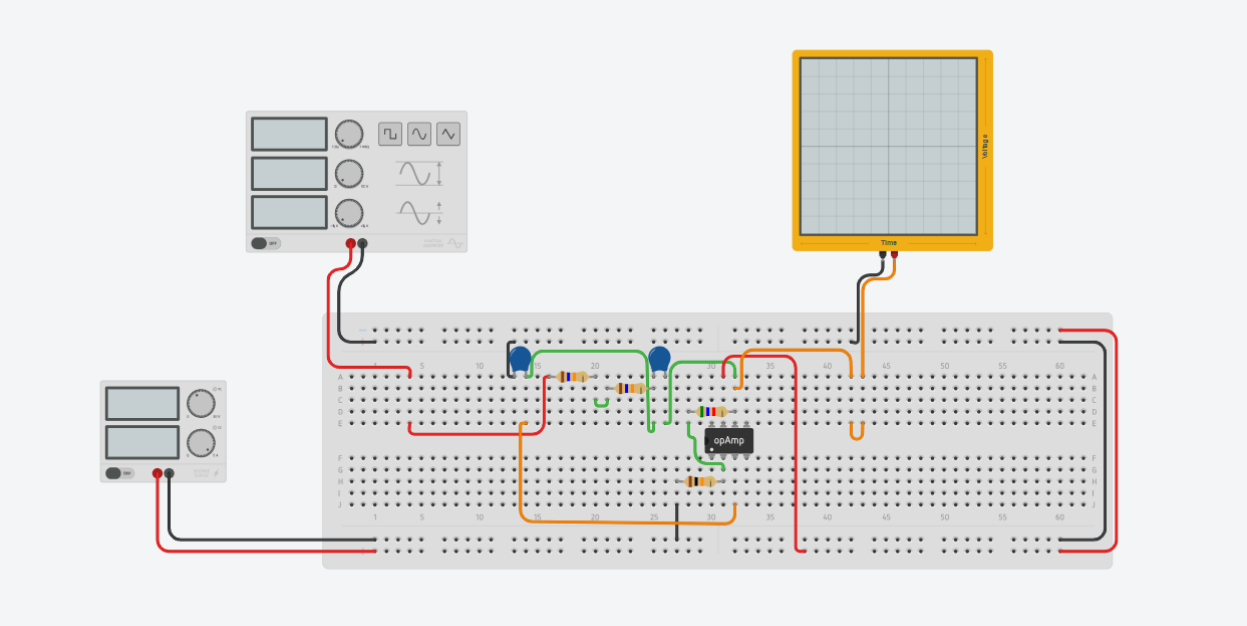
Design, simulate, and build a second-order Butterworth low-pass filter with f0 ≈ 1 kHz, and characterize its frequency response across all five platforms.

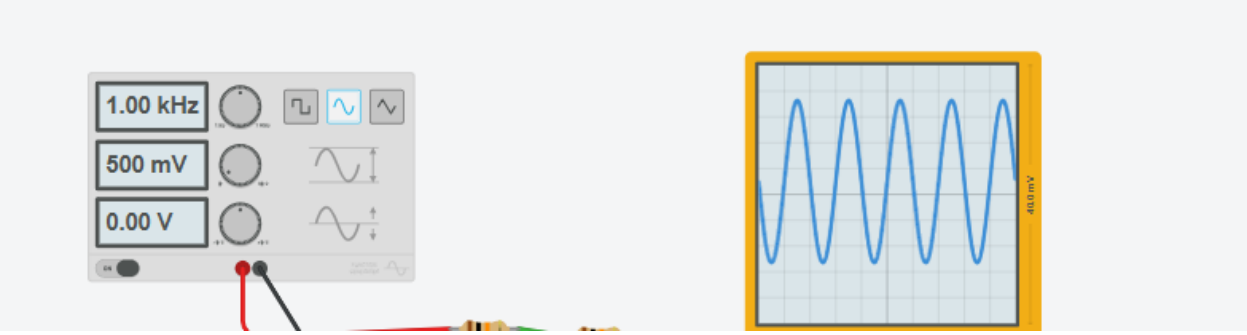
4.2 Circuit Diagram



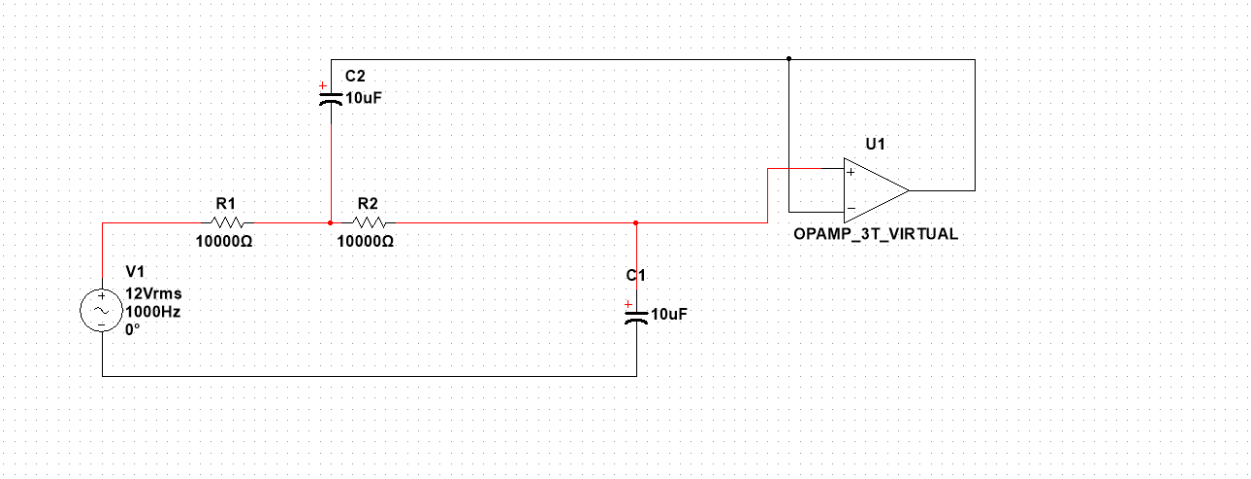
4.3 Procedure

Tinkercad





Multisim



MATLAB

```
R = 16e3; C = 10e-9; Rf = 5.6e3; Rg = 10e3;

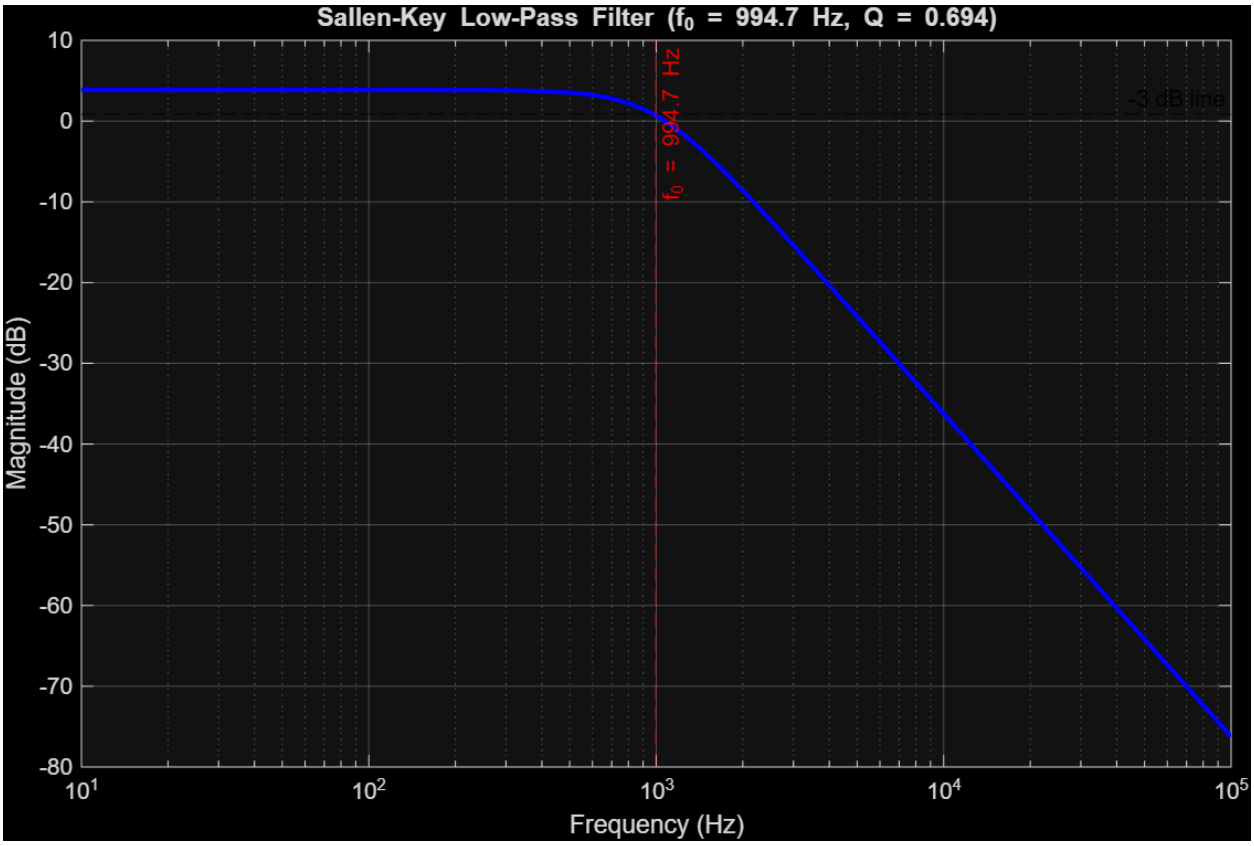
K_lpf  = 1 + Rf/Rg;

w0_lpf = 1/(R*C);

f0_lpf = w0_lpf/(2*pi);

Q_lpf  = 1/(3 - K_lpf);

H_lpf  = tf(K_lpf*w0_lpf^2, [1, w0_lpf/Q_lpf, w0_lpf^2]);
```


Sallen-Key Low-Pass Filter (f_0 = 994.7 Hz, Q = 0.694)
10
0
-10
-20
-30
-40
-50
-60
-70
-80
Magnitude (dB)
10^1
10^2
10^3
10^4
10^5
Frequency (Hz)
f_0 = 994.7 Hz
-3 dB line

Hardware

1. Build the circuit on a breadboard; add 0.1 µF decoupling capacitors at the op-amp supply pins.
2. Apply a sine wave from the function generator; sweep frequency per Table 4.4a.
3. Record V_{in}, V_{out}, and phase shift (scope cursors or Lissajous method) at each frequency.
4. Identify the measured −3 dB cutoff and compare to theory.

4.4 Data Tables

(a) Frequency Response — record at each platform

Frequency (Hz)	Vin (V)	Vout — Theory	Vout — Tinkercad	Vout — Multisim	Vout — Hardware	Phase (°)

(b) Summary — Measured vs. Calculated Cutoff

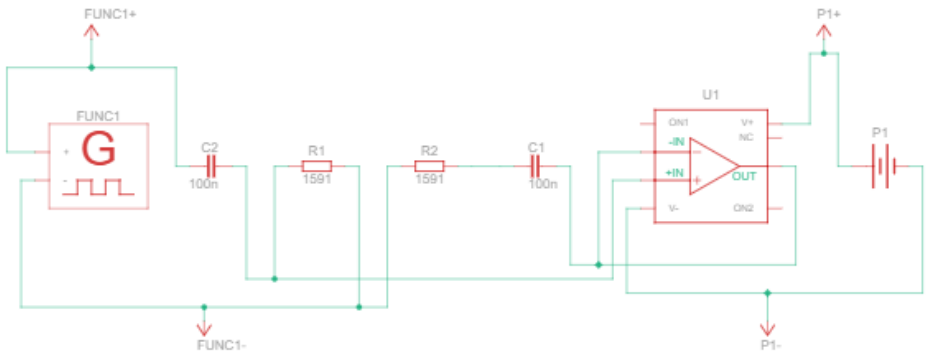
Source	Measured/Simulated f0 (Hz)	% Error vs. Calculated

5. Experiment 2 — Sallen-Key High-Pass Filter

5.1 Objective

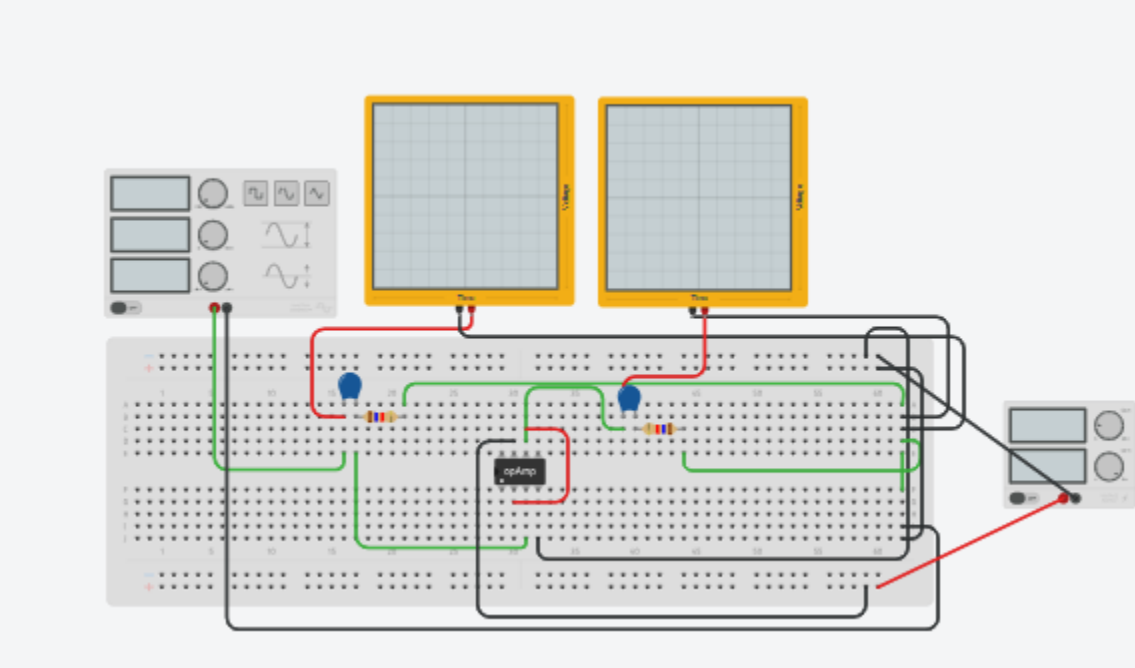
Repeat the Experiment 1 procedure for the high-pass configuration (R and C positions swapped), f0 ≈ 1 kHz.

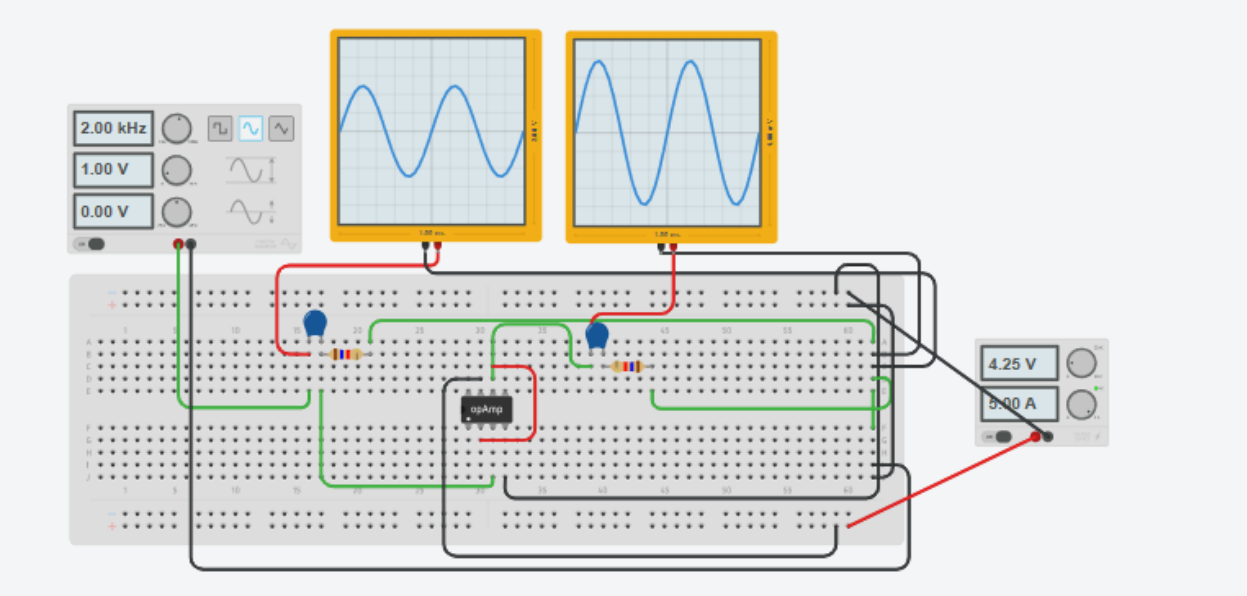
5.2 Circuit Diagram



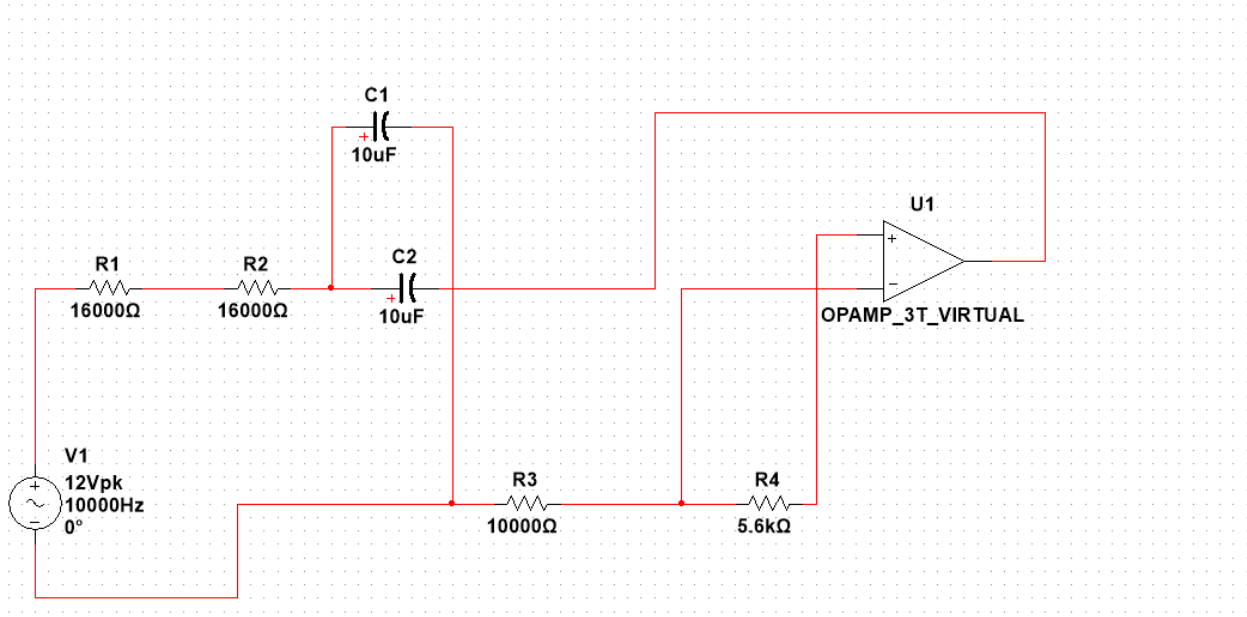
5.3 Procedure

Tinkercad



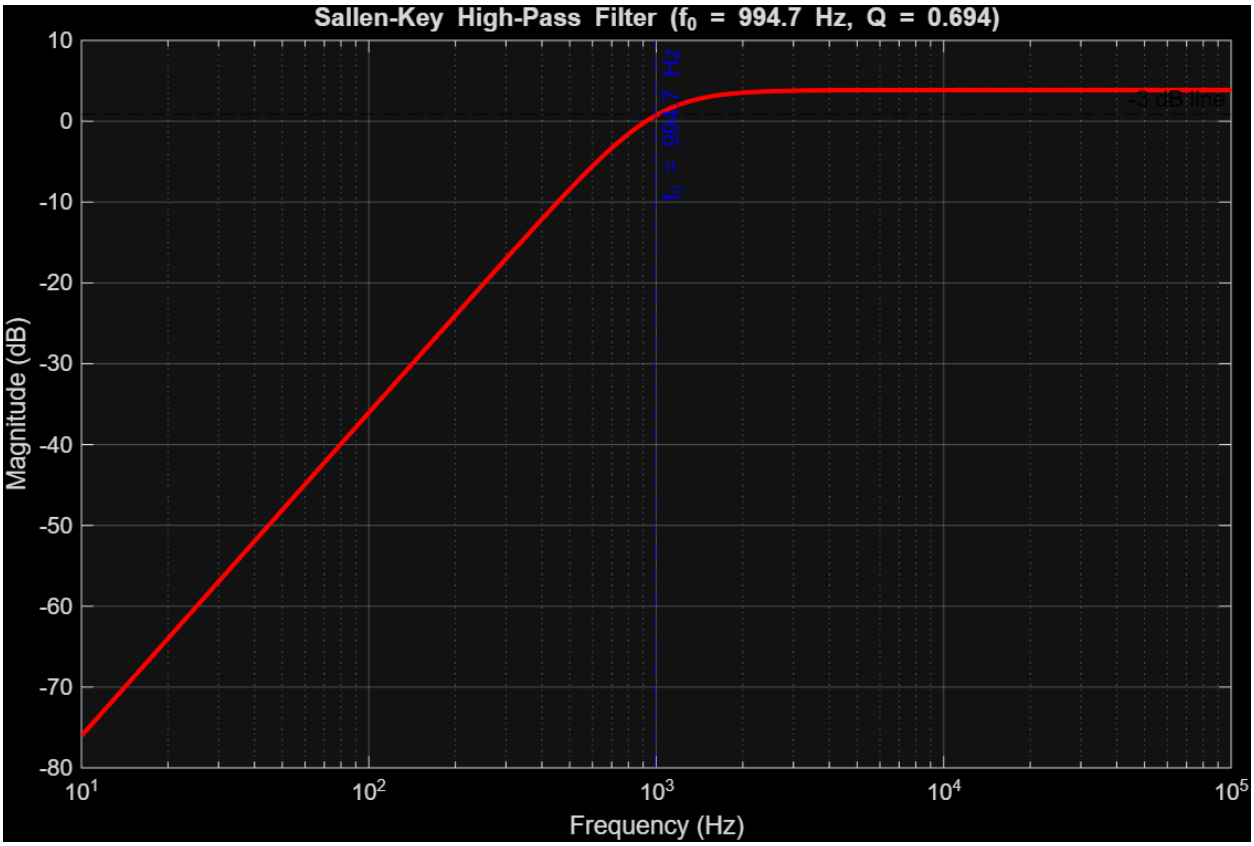


MultiSim



MatLab

```
K_hpf  = K_lpf;
w0_hpf = w0_lpf;
f0_hpf = f0_lpf;
Q_hpf  = Q_lpf;
H_hpf  = tf(K_hpf*[1 0 0], [1, w0_hpf/Q_hpf, w0_hpf^2]);
```



5.4 Data Tables

Frequency (Hz)	Vin (V)	Vout — Theory	Vout — Tinkercad	Vout — Multisim	Vout — Hardware	Phase (°)

Frequency (Hz)	Vin (V)	Vout — Theory	Vout — Tinkercad	Vout — Multisim	Vout — Hardware	Phase (°)

Source	Measured/Simulated f0 (Hz)	% Error vs. Calculated

6. Experiment 3 — Band-Pass Filter

6.1 Objective

Cascade the Experiment 2 high-pass stage (f_low ≈ 300 Hz) with a second low-pass stage (f_high ≈ 3 kHz) to realize a band-pass response, and characterize center frequency, bandwidth, and passband gain.

6.2 Circuit Diagram

[Insert / sketch the two-stage cascade: HPF stage output feeds directly into the LPF stage input. Both stages share supply rails; each has its own decoupling capacitors.]

6.3 Procedure

Repeat the five-platform procedure from Section 4.3 for the cascaded circuit, using the component values in Section 3.3. Sweep frequency across the full range (e.g., 50 Hz – 10 kHz) to capture both roll-offs and the passband plateau.

6.4 Data Tables

Frequency (Hz)	Vin (V)	Vout — Theory	Vout — Tinkercad	Vout — Multisim	Vout — Hardware	Phase (°)

Bandwidth Summary

Source	f_low (Hz)	f_high (Hz)	Bandwidth (Hz)	Center f (Hz)

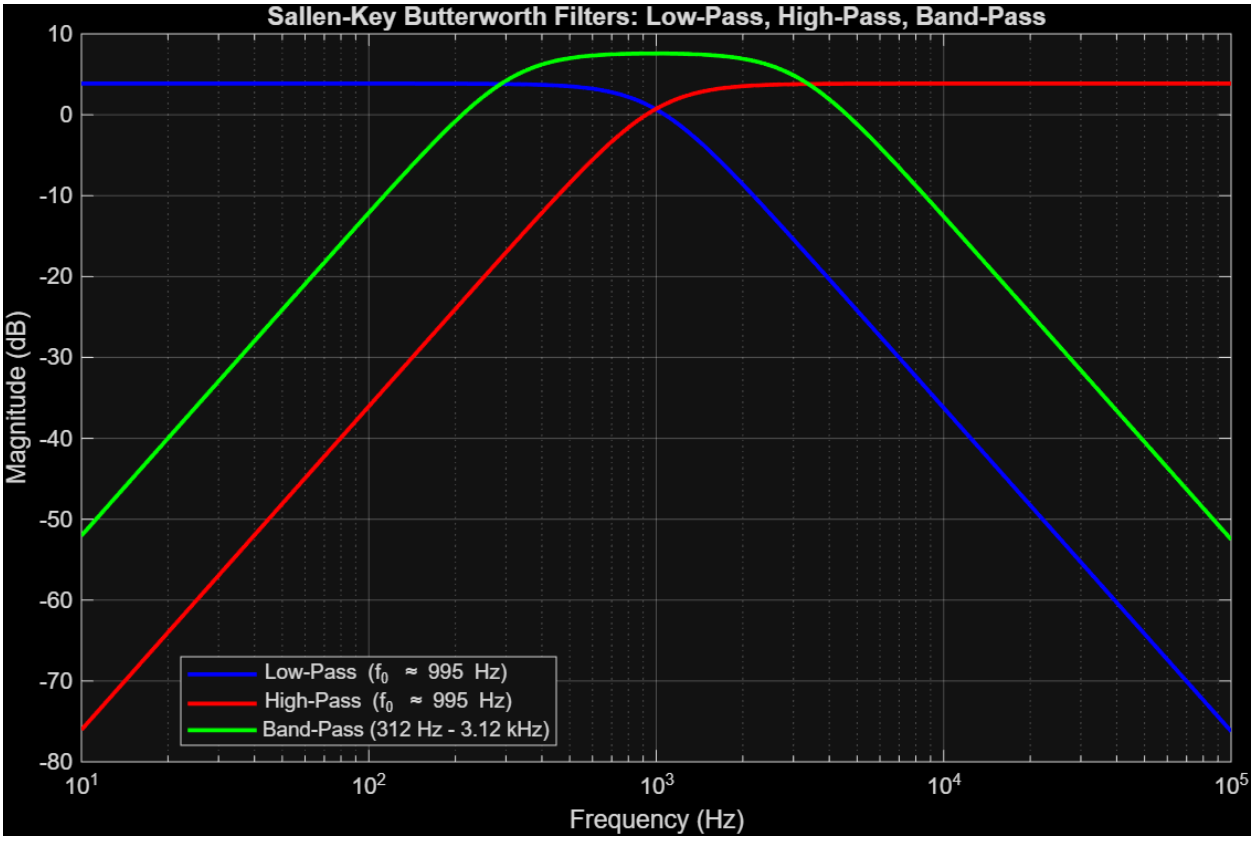
6.5 Analysis Questions

5. How does the measured bandwidth compare to (f_high − f_low) predicted from the individual stage cutoffs?
6. Does cascading two stages reduce the overall passband gain compared to a single stage? Explain using the individual stage gains K.

7. Comparative Analysis: Theory vs. Simulation vs. Hardware

Using the data collected in Sections 4–6, overlay the Bode magnitude (and phase) curves from MATLAB (theoretical reference), and hardware measurements on a single plot for each filter type. Paste or embed the combined plots below.

7.1 Combined Bode Plot



8. Error Analysis

- Component tolerance: standard 5% resistors/capacitors shift f0 and Q from nominal — quantify using the rounded values from Section 3 vs. the exact design targets.
- Op-amp non-idealities: gain-bandwidth product and slew rate limit high-frequency accuracy — identify at which test frequency (if any) this became visible.
- Breadboard parasitics: stray capacitance and lead inductance can shift response at higher frequencies or cause instability/oscillation.
- Measurement uncertainty: oscilloscope reading resolution, probe loading, and function generator amplitude drift.

Summarize the dominant error source(s) for each filter type and estimate their approximate contribution to the observed deviation between calculated and measured f0.

9. Conclusion

[Summarize what was designed, simulated, and built; state whether the measured cutoff frequencies, gain, and phase matched theoretical predictions within acceptable tolerance; note the most significant real-world limitation observed (e.g., slew-rate distortion, component tolerance) and what you would change in a redesign.]

Appendix A — MATLAB Reference Script

```
R = 16e3; C = 10e-9;          % low-pass stage
wn = 1/(R*C);                 % rad/s
Q = 0.707;
H = tf([wn^2], [1 wn/Q wn^2]);
figure; bode(H); grid on;
[mag, phase, w] = bode(H, 2*pi*logspace(1,4,200));
f0_actual = wn/(2*pi)         % compare to hardware-measured -3dB point
```

Appendix B — Safety Notes

- Verify supply polarity before powering the op-amp — reversed ±12 V can destroy the device instantly.
- Always place 0.1 µF decoupling capacitors as close as possible to the op-amp supply pins.
- Do not exceed the op-amp's rated supply voltage or output current.
- Power down the circuit before making any wiring changes on the breadboard.